DSC 40A - Homework 2

due Tuesday, July 7th at 11:59 PM

Homeworks are due to Gradescope by 11:59PM on the due date.

You may request an extension for this assignment by submitting the form.

Homework will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically. You should **always explain and justify** your conclusions, using sound reasoning. Your goal should be to convince the reader of your assertions. If a question does not require explanation, it will be explicitly stated.

Homeworks should be written up and turned in by each student individually. You may talk to other students in the class about the problems and discuss solution strategies, but you should not share any written communication and you should not check answers with classmates. You can tell someone how to do a homework problem, but you cannot show them how to do it. **Only handwritten solutions will be accepted (use of tablets is permitted). Do not typeset your homework (using LaTeXor any other software)**. Staff will not grade typed submissions.

For each problem you submit, you should **cite your sources** by including a list of names of other students with whom you discussed the problem. Instructors do not need to be cited.

This homework will be graded out of **33 points**. The point value of each problem or sub-problem is indicated by the number of avocados shown.

Problem 1. Reflection and Feedback Form

Make sure to fill out this Reflection and Feedback Form, linked here, for two points on this homework! This form is primarily for your benefit; research shows that reflecting and summarizing knowledge helps you understand and remember it.

Problem 2. Command+C Command+V

Consider a dataset of n numbers $y_1, y_2, ..., y_n$ with mean M and standard deviation S. Suppose we introduce k new values to the dataset, $y_{n+1}, y_{n+2}, ..., y_{n+k}$, all of which are equal to M.

Let the new mean and standard deviation of all $n+k$ values be M' and S', respectively.

a) Find M' in terms of M, n, k, and S. (You may not need to use all of these variables in your answer.)

b) Find S' in terms of M, n, k, and S. (You may not need to use all of these variables in your answer.)

Problem 3. Slippery Slope

As you learned from the lectures, $h^* = \text{Median}(y_1, y_2, ..., y_n)$ is the constant prediction that minimizes mean absolute error:

$$R_{\text{abs}}(h) = \frac{1}{n}\sum_{i=1}^{n} |y_i - h|$$

Suppose that we have a dataset of numbers $y_1, y_2, ..., y_n$ such that n **is odd**, all values y_i are **distinct**, and that the values are arranged in increasing order. That is, $y_1 \leq y_2, \leq \ldots \leq y_n$.

Note: Parts (a) and (b) are independent of each other.

a) Suppose that $R_{\text{abs}}(\alpha) = V$, where V is the minimum value of $R_{\text{abs}}(h)$ and α is one of the numbers in our dataset.

Let $\alpha + \beta$ be the smallest value greater than α in our dataset, where $\beta > 0$. Another way of thinking about this is that $\beta = (\text{smallest value greater than } \alpha) - \alpha$.

Suppose we modify our dataset by replacing the value α with the value $\alpha + \beta + 1$. In our new dataset of n values, what is the new minimum value of $R_{\text{abs}}(h)$ and at what value of h is it minimized? Your answers to both parts should only involve the variables V, α, β, n, and/or one or more constants.

b) Let y_a and y_b be two values in our dataset such that $y_a < y_b$ and that the slope of $R_{\text{abs}}(h)$ is the same between $h = y_a$ and $h = y_b$. Specifically, let d be the slope of $R_{\text{abs}}(h)$ between y_a and y_b.

Suppose we introduce a new value q to our dataset such that $q > y_b$. In our new dataset of $n+1$ values, the slope of $R_{\text{abs}}(h)$ is still the same between $h = y_a$ and $h = y_b$, but it's no longer equal to d. What is the slope of $R_{\text{abs}}(h)$ between $h = y_a$ and $h = y_b$ in our new dataset? Your answer should depend on d, n, q, and/or one or more constants.

Problem 4. Pop Quiz

Complete the following two concept checks.

a) Determine whether each statement is true or false. Explain briefly.

(a) If the training features $x_1, \ldots, x_n$ for a simple linear regression problem have mean zero then the intercept of the optimal linear model will equal zero.

(b) The line of best fit is found by minimizing the empirical risk.

(c) The slope of the line of best fit is always positive.

(d) A simple linear regression model has exactly two parameters.

b) Match each term (a)–(e) with its correct definition (1)–(5) (no explanation required):

(a)	Empirical risk	Equations obtained by setting partial derivatives of empirical risk to zero.	(1)
(b)	Intercept	Average value of the loss function over the training data.	(2)
(c)	Square loss	Parameter indicating the amount that the predicted value changes if the feature increases by one unit.	(3)
(d)	Normal equations	Parameter indicating the predicted value when the feature is zero.	(4)
(e)	Slope	A loss function measuring the squared difference between prediction and actual value.	(5)

Problem 5. Simple Linear Regression

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

a) For the data above, explain in words how the scatterplot (x on horizontal axis and y on vertical) looks like.

b) Find the best fitting line using linear regression with x as predictor and y response. What is the intercept and slope?

c) Now, find the best fitting line using linear regression with x^2 as predictor and y response. What is the intercept and slope?

d) Now that we are using x^2 as feature, explain in words how the scatterplot (x on horizontal axis and y on vertical) looks like.

e) Were the slopes calculated in (b) and (c) different from each other? If yes, please explain the reason for the change.

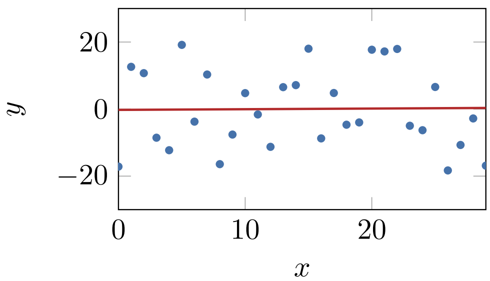
Problem 6. Explain The Plot

The four plots below contain a training dataset $\{(x_i, y_i)\}_{i=1}^{30}$ of thirty points in the xy-plane (blue), along with the line of best fit for the associated simple linear regression problem (red).

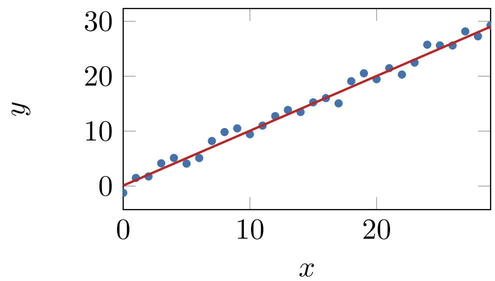
a) 🥑🥑 For each plot below, determine whether the simple linear model is appropriate. Write a short explanation of your reasoning.

b) 🥑🥑🥑 For each plot, consider the value $R(w_0^*, w_1^*)$ of the empirical risk associated to the optimal parameters for the simple linear model and given training dataset. Rank the plots from least to greatest value of $R(w_0^*, w_1^*)$.

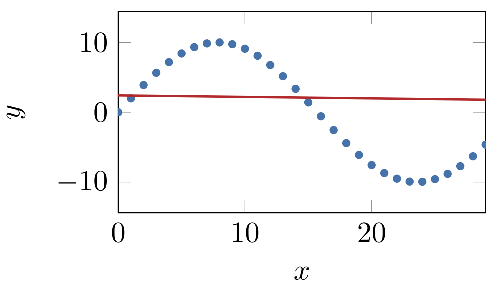
(i)



(ii)



(iii)



(iv)